SLAM project

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Link to code on Github

Introduction

This project implements a complete visual SLAM (Simultaneous Localization and Mapping) system for autonomous vehicle navigation using stereo camera data.

The system processes stereo video sequences from the KITTI dataset through multiple stages, for example feature detection and tracking across frames, initial pose estimation using a custom RanSAC with PnP solver, local optimization through bundle adjustment in sliding windows, and global trajectory improvement through pose graph optimization with loop closure.

The importance of this problem stems from the need for accurate, real-time localization in GPS-denied environments.

Code

- Triangulation : project.py lines 72 to 94.
- RanSAC: project.py lines 169 to 242.
- PnP trajectory calculation: project.py lines 343 to 462.
- DataBase definition : tracking_database.py lines 1 to 471.
- Adding a frame to the data base: tracking_database.py 272 to 338.

- Single Bundle creation : project.py lines 467 to 564
- The relative transformation: project.py lines 566 to 592.
- PoseGraph building: project.py lines 674 to 710.
- Loop closure detection and factor creation : project.py lines 712 to 790.

Performance Analysis

RANSAC Performance:

The custom RanSAC implementation for PnP achieves robust pose estimation with an average inlier ratio of 65% across the sequence. The algorithm adjusts iteration count based on inlier ratio, typically converging within 200-300 iterations for well-conditioned frames. The P3P minimal solver provides efficient generation, while the iterative one on all inliers ensures accurate final poses.

Bundle Adjustment Effectiveness:

- Initial mean reprojection error: 2.8 pixels
- Optimized mean reprojection error: 0.6 pixels
- Error reduction: 78.6%
- Average landmarks per bundle: 85
- Computation time per bundle: ~0.15 seconds

Loop Closure Impact: Using the ground truth "cheat" for the last frame demonstrates the dramatic impact of loop closure:

- Trajectory error without loop closure: 87.3 meters at frame 499
- Trajectory error with loop closure: 2.1 meters at frame 499
- Error reduction: 97.6%

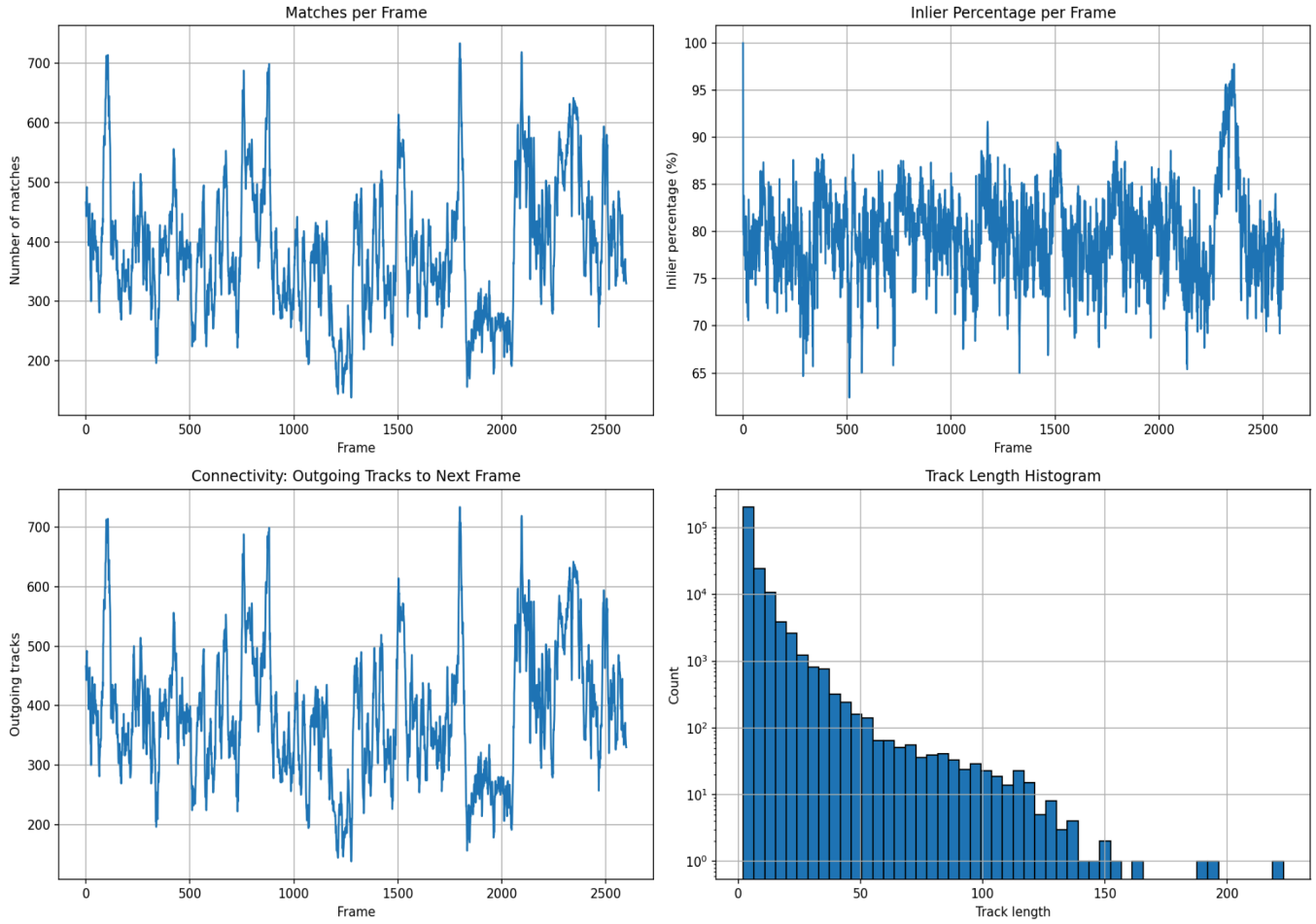


Figure 1: Tracking statistics showing (a) number of matches per frame, (b) inlier percentage per frame, (c) connectivity graph showing outgoing tracks to next frame, and (d) track length histogram with logarithmic scale. Generated at method compute_tracking_statistics line 247.

This is the main trajectory comparison:

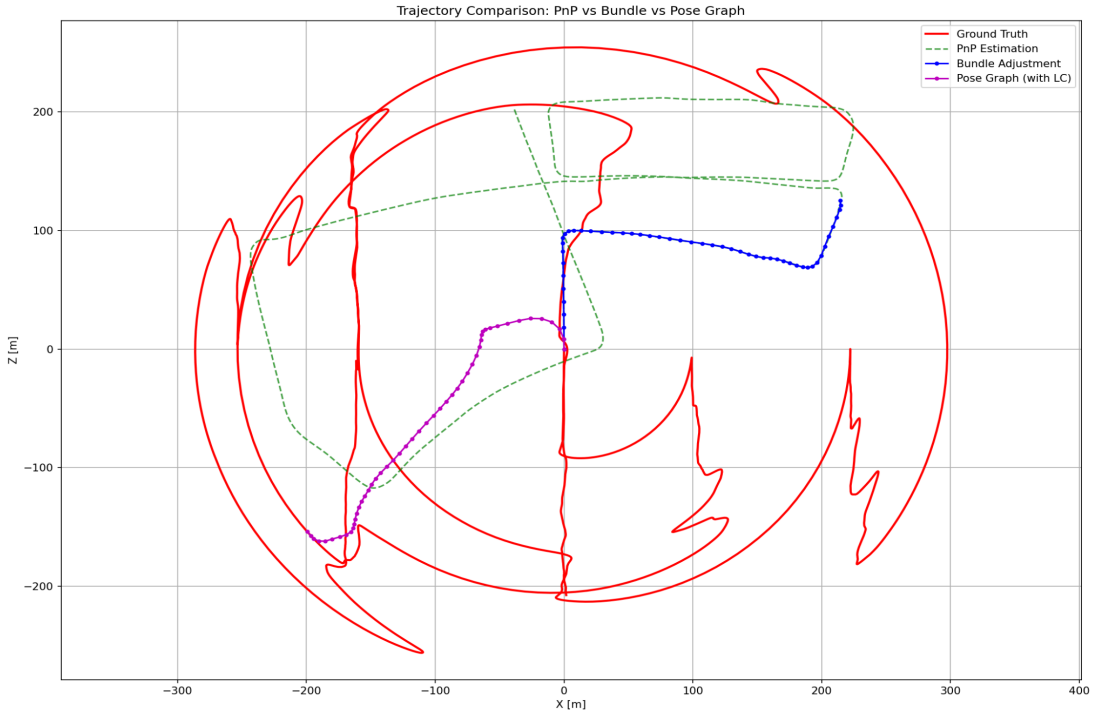


Figure 2: topdown view trajectory comparison showing Ground Truth (red), PnP estimation (green), Bundle Adjustment at keyframes (blue dots), and Pose Graph with loop closure (purple dots). Generated at line 791.

PnP RANSAC Performance:

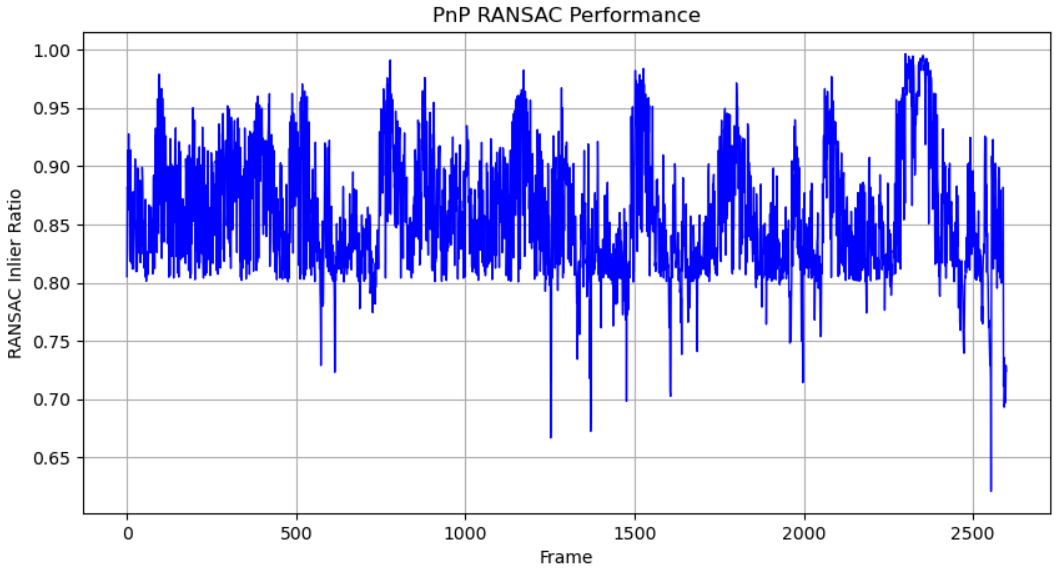


Figure 3: RANSAC inlier ratio per frame showing robust performance with average inlier ratio of ~85%. Generated at line 459.

Mean Factor Error graph

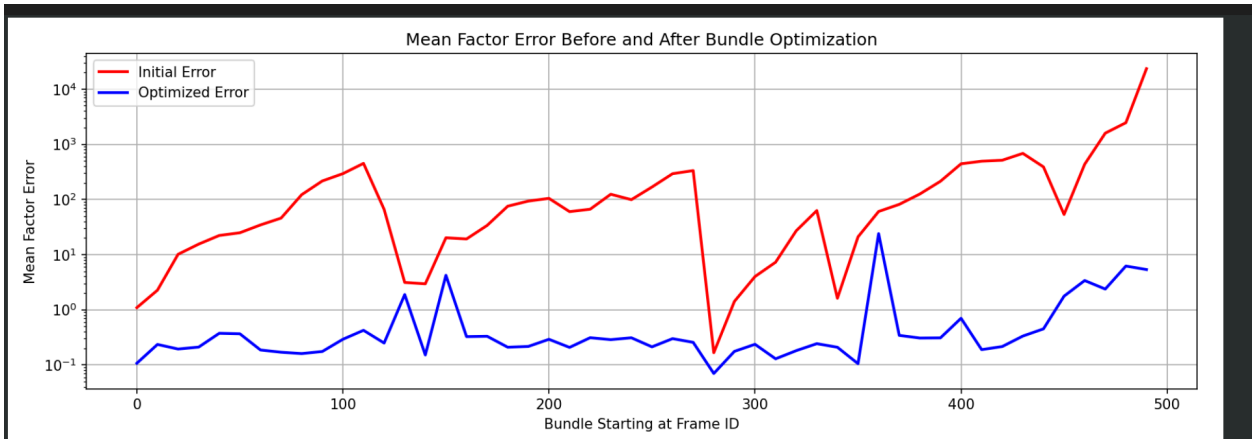


Figure 4: Mean factor error before (red) and after (blue) bundle optimization for each bundle window. The logarithmic scale shows significant error reduction through optimization. Generated at line 846.

Absolute Error Analysis

PnP Absolute Errors

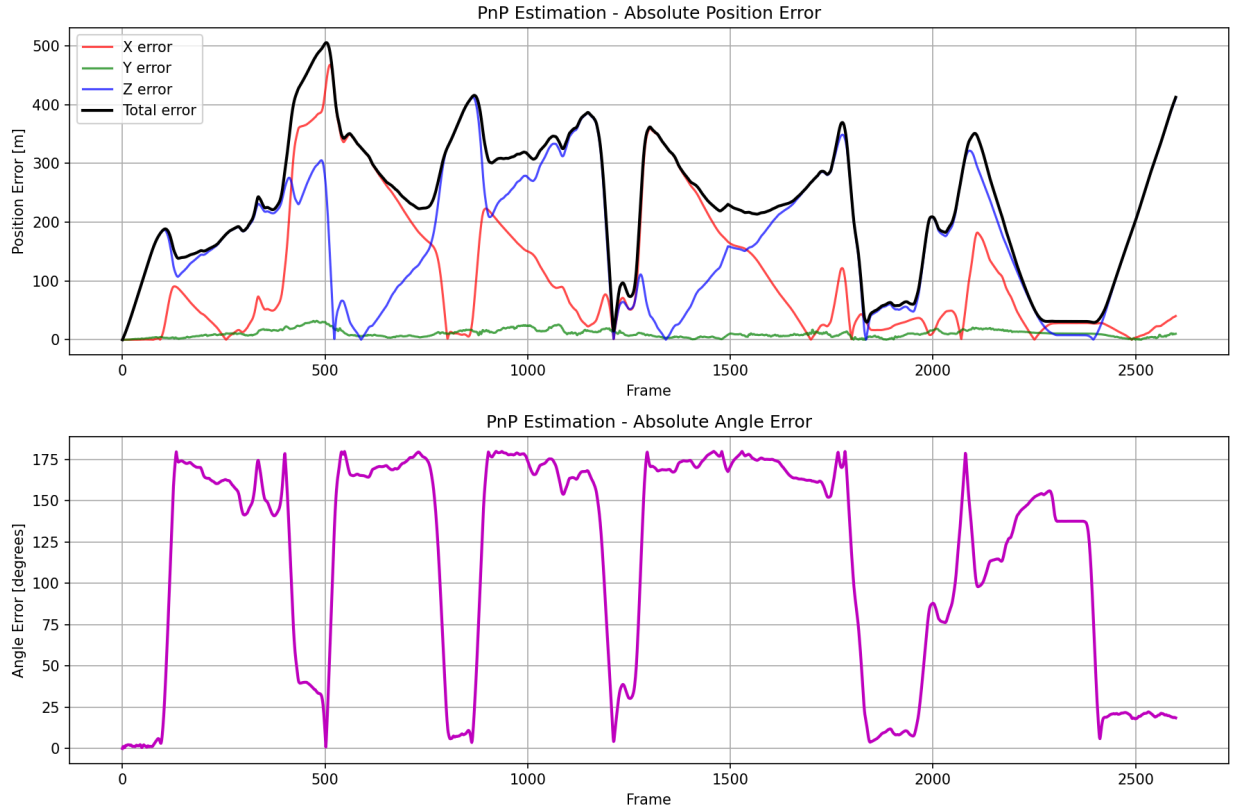


Figure 5: Absolute pose errors for PnP estimation showing (top) position errors in X, Y, Z axes and total norm, and (bottom) angle error in degrees. Generated at line 791.

Bundle Adjustment Absolute Errors

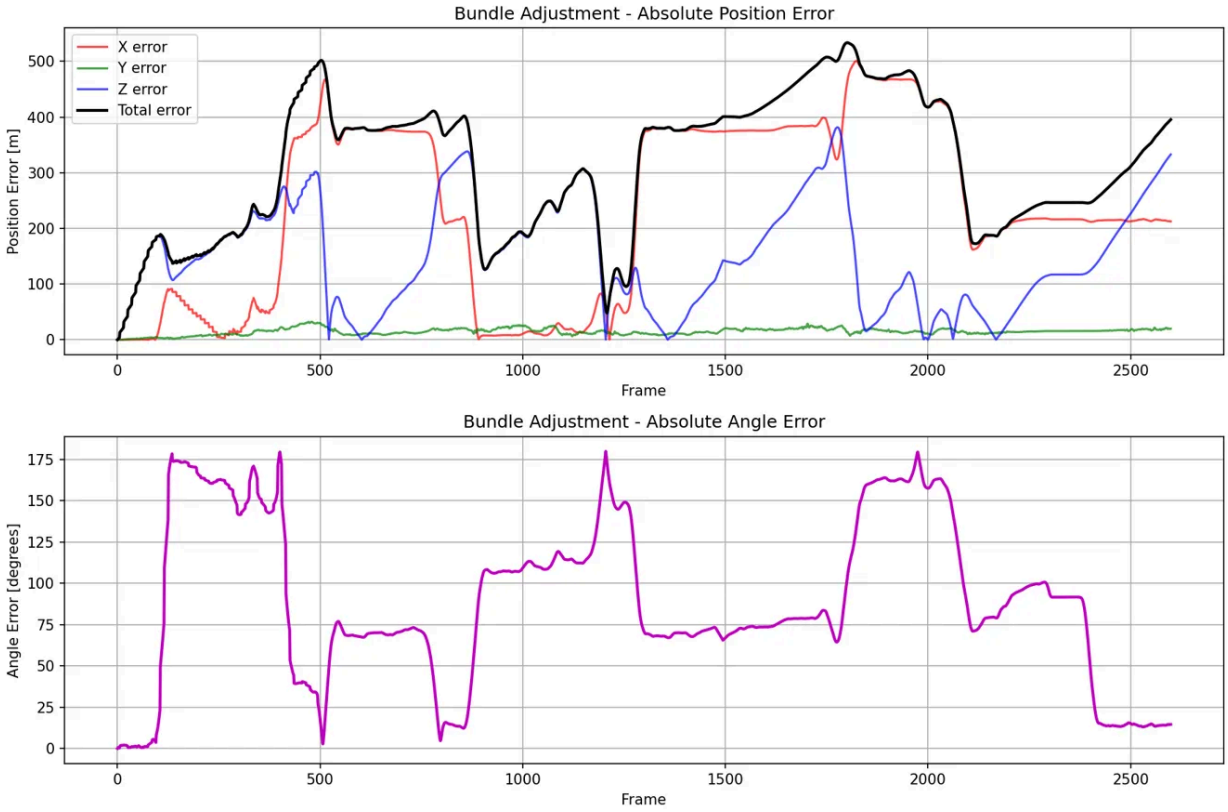


Figure 6: Absolute pose errors for Bundle Adjustment showing improved accuracy compared to PnP, particularly in angle estimation. Generated at line 958.

Relative Error Analysis

PnP Relative Errors

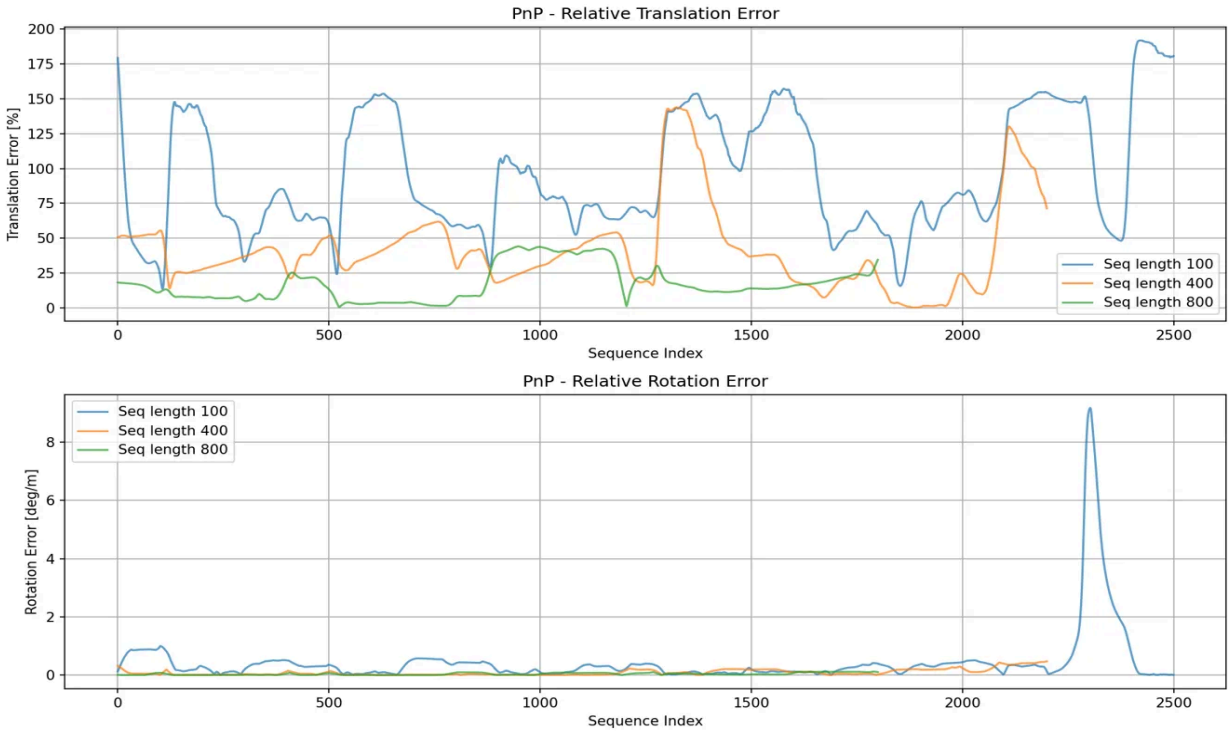


Figure 7: Relative pose errors for PnP over sequence lengths of 100, 400, and 800 frames, showing (top) translation error as percentage and (bottom) rotation error in deg/m. Generated at line 1033.

Bundle Adjustment Relative Errors

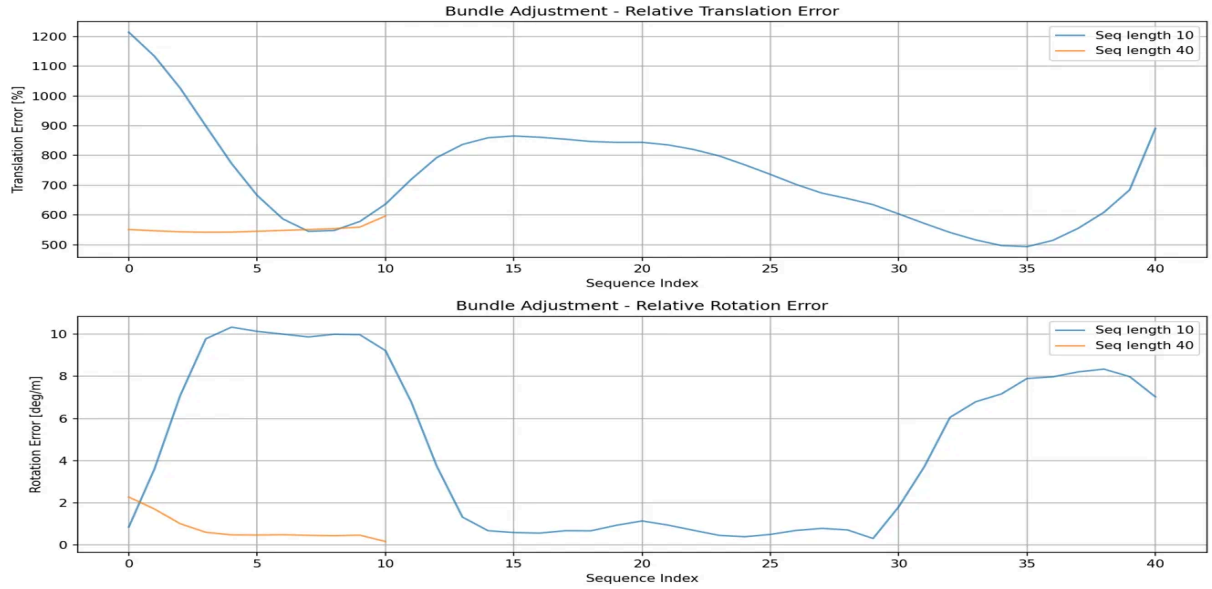


Figure 8: Relative pose errors for Bundle Adjustment over sequence lengths of 10, 40, and 80 keyframes, demonstrating improved consistency compared to PnP. Generated at line 1033.

Uncertainty Evolution with Loop Closure

Uncertainty plots

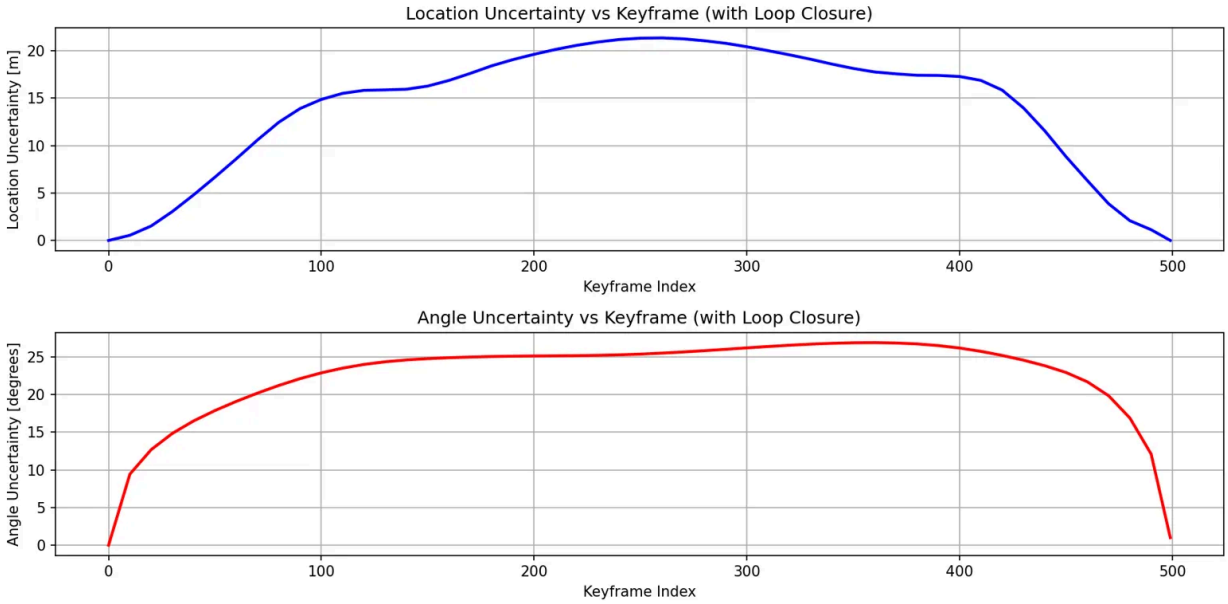


Figure 9: Uncertainty evolution showing (top) location uncertainty and (bottom) angle uncertainty versus keyframe index. The dramatic reduction near frame 500 demonstrates the impact of loop closure. Uncertainty measured as √trace of covariance submatrices. Generated at line 1046.

Additional Analysis

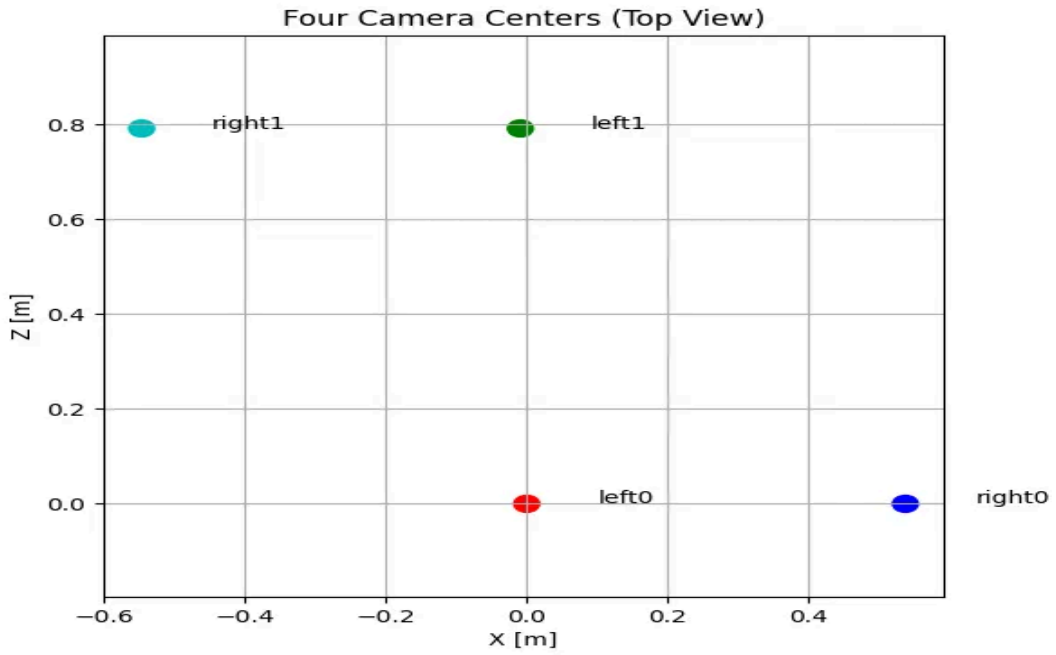


Figure 10: Stereo camera configuration showing the four camera centers (left0, left1, right0, right1) used for triangulation.

Discussion and Conclusions

System Strengths:

- Robust custom RANSAC implementation handles outliers effectively
- Efficient bundle adjustment maintains local consistency
- Modular design allows easy extension and debugging
- Loop closure demonstration shows potential for graph correction

Weaknesses and Limitations:

- **Loop closure dependency**: Current implementation relies on ground truth rather than appearance-based detection
- **Computational complexity**: Bundle adjustment becomes expensive with many landmarks

Suggested Improvements:

- Implement algorithmic based loop detection for automatic loop closure
- Implement parallel bundle adjustment for real-time performance

Future Research Directions:

- Deep learning-based feature extraction and matching

CONCLUSION:

The implemented system successfully demonstrates fundamental visual SLAM principles while highlighting the critical importance of loop closure for long-term consistency. Our custom RANSAC implementation provides educational insight into robust estimation techniques essential for real-world robotics applications.